Decoding Human Behavior Through CAPTCHA Interaction Patterns

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Summary

Can we determine human interaction styles in CAPTCHA systems?

User behavior varies widely: from fast, impulsive actions to careful, deliberate movements. To capture this, we analyze 30,000 mouse trajectory sessions, adjusting movement features based on task context to uncover distinct behavioral profiles.We then enable real-time classification, allowing CAPTCHA systems to identify user behavior types and improve decision-making.

Data Overview

We use data collected by CapyCap,Inc:

- 30,000 Total sessions,
- Data normalized to fit 200 × 200 canvas
- 3 CAPTCHA game types:

Sheep Herding

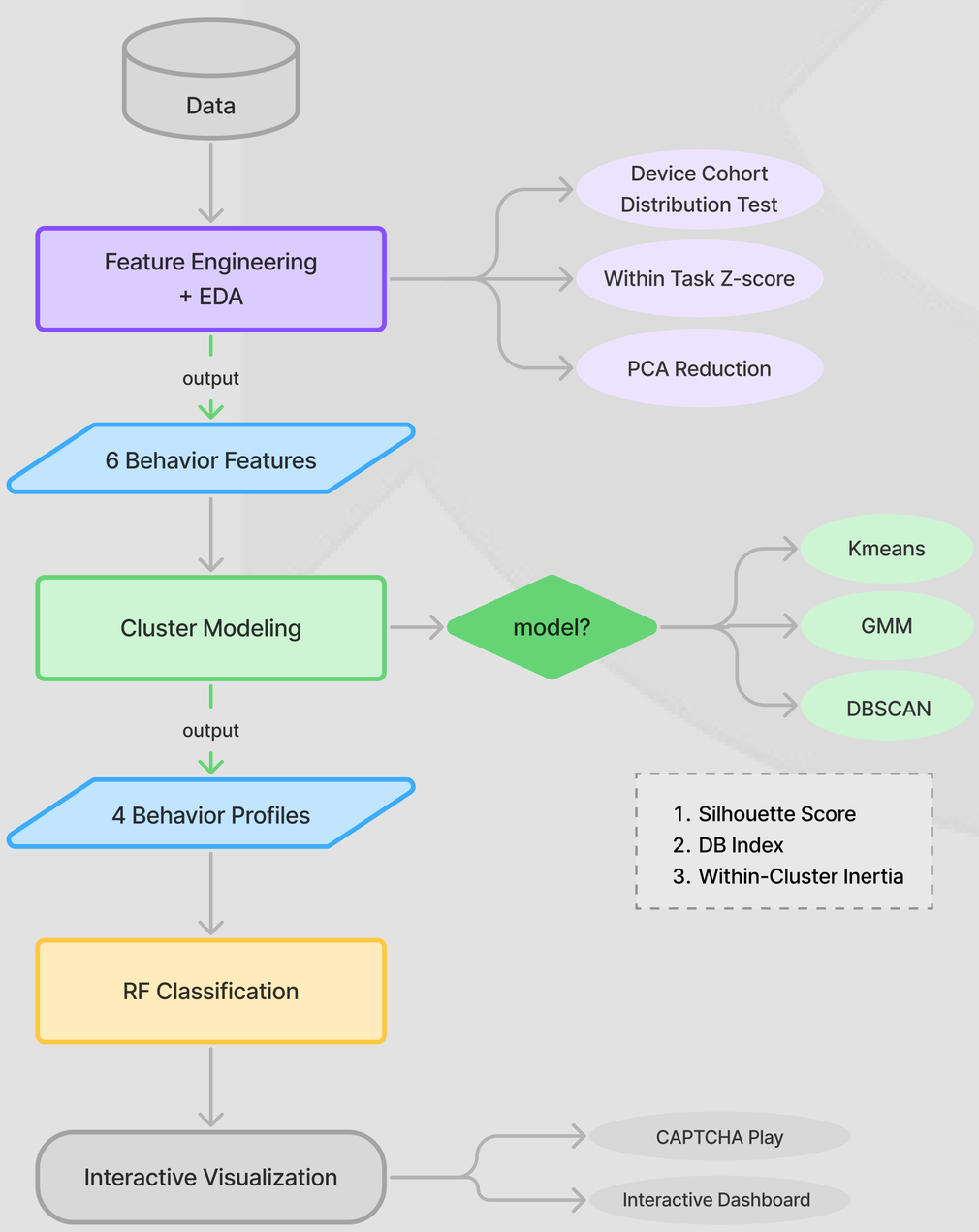
Thread the Needle

Polygon Stacking

On top of the collected data, we extracted 6 behavior features:

Feature	What it captures
duration	Total session time (ms)
path_length	Total cursor distance (px)
speed_mean	Avg. step distance (px/sample)
path_efficiency	Straight-line ÷ path length (1 = direct)
pause_rate	Fraction of near-stationary steps
speed_std	Speed variability (bursts vs. smooth)

Approach



Evaluation/ Result

Feature Engineering

- Touch and non-touch device cohorts exhibit significantly different behavioral distributions, so touch sessions were excluded.
- Features were normalized within each game type to remove difficulty-driven baselines, ensuring behavior is evaluated relative to task context.
- After PCA, the feature space was reduced to four principal components for efficient modeling.

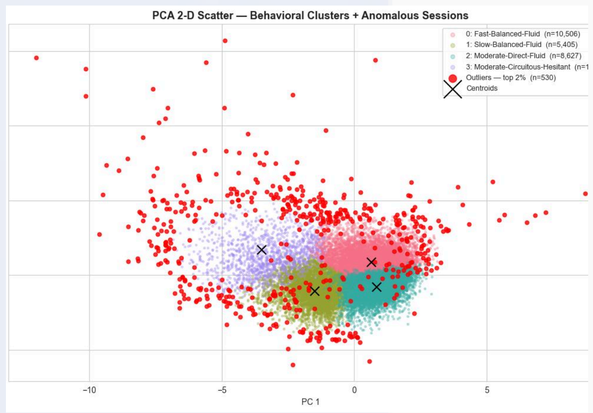
KMeans model selection

KMeans was selected for its interpretability in identifying behavior groups. K = 4 determined based on 3 measures (elbow, silhouette, Davies–Bouldin) by median voting across their optimal values .

Profile Group Identification

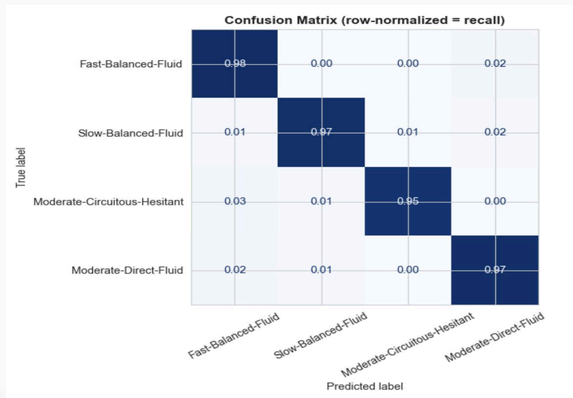
Four distinct behavior groups were identified based on speed, efficiency, and hesitation:

- Fast–Balanced–Fluid (n = 10628)
- Slow–Moderate–Fluid (n = 5471)
- Moderate–Direct–Fluid (n = 8686)
- Moderate–Circuitous–Hesitant (n = 1684)



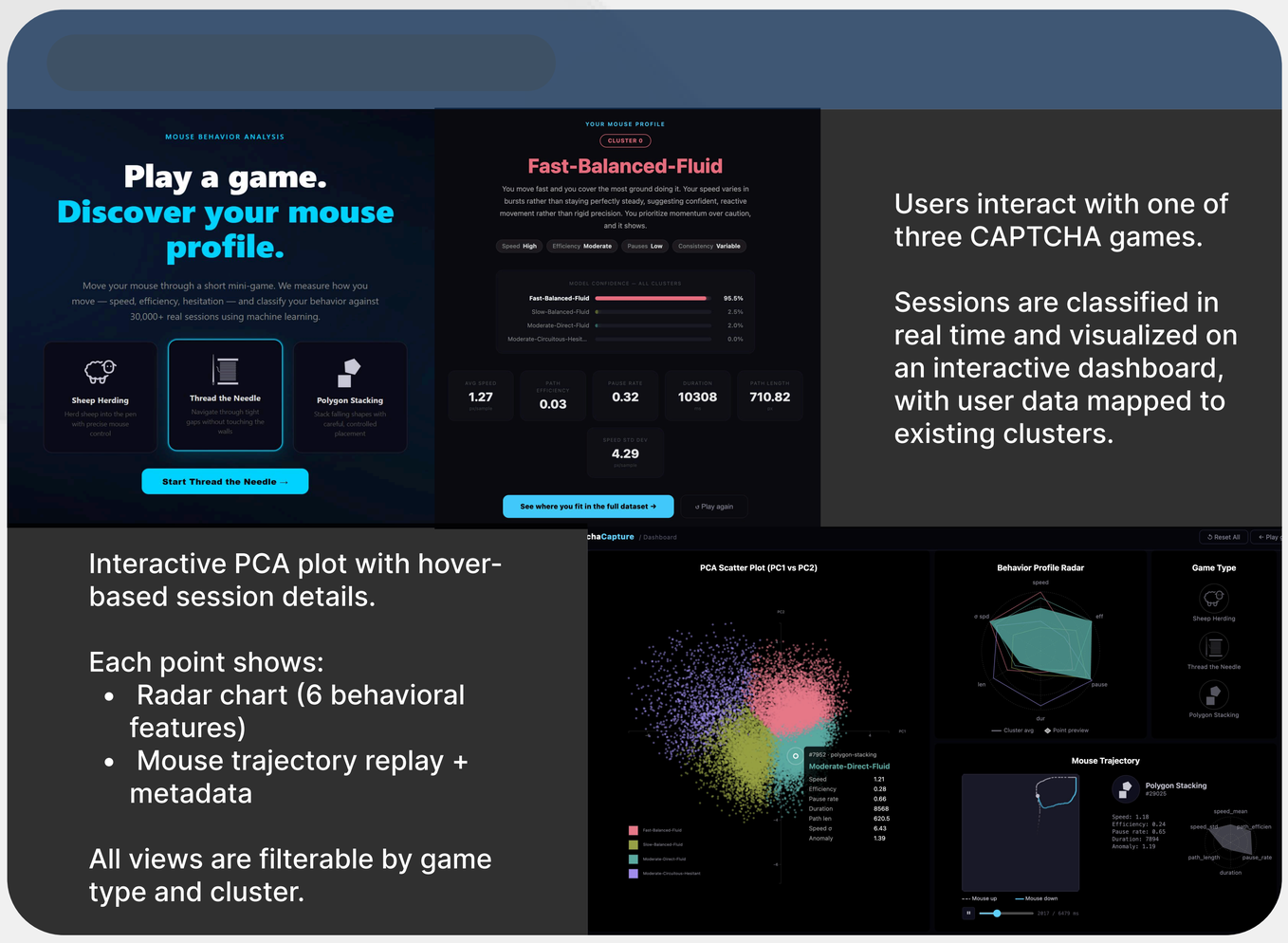
Top 2% (n = 530) were flagged as anomalies, showing longer paths and higher pause rates, suggesting potential bot activity.

Random Forest Classification



A Random Forest was trained on normalized behavioral features to predict cluster membership, using macro F1-score to address class imbalance.

Achieved 97.0% accuracy (macro F1 = 0.967), with stable cross-validation performance (0.969 ± 0.004)
Path efficiency is the most informative feature, followed by speed mean.



Users interact with one of three CAPTCHA games.

Sessions are classified in real time and visualized on an interactive dashboard, with user data mapped to existing clusters.

Interactive PCA plot with hover-based session details.

Each point shows:

- Radar chart (6 behavioral features)
- Mouse trajectory replay + metadata

All views are filterable by game type and cluster.